Emotional Adaptive Navigation for Shopify — Planning Document

Research cut-off: 21 October 2026
Scope: Product, evidence, platform, legal/compliance, technical, evaluation, and business planning. This is not legal advice and contains no implementation code.

Citation convention: [PAPER] refers to one of the three supplied PDFs; **[OUTSIDE]** refers to independent legal, scientific, platform, or competitor sources; **[INFERENCE]** marks this document's recommendation or interpretation rather than a directly demonstrated result. Legal/platform links are dated or marked with the access date in the source list.

Executive summary

The product is **Emotional Adaptive Navigation for Shopify**: a storefront app that estimates a shopper's likely affective state—such as frustration, confusion, uncertainty, engagement, or confidence—from non-biometric interaction signals, then adapts its own navigation and product-discovery module. The estimate must be probabilistic and reversible, never presented as a diagnosis. The supplied papers do not test emotion inference: they neither establish its effectiveness nor refute it. Independent studies provide preliminary, mixed evidence that keyboard, touch, cursor, and browsing patterns can carry affective information, but most are lab, task-specific, small, and not validated on commerce storefronts.

The defensible build is **privacy-first Emotional Adaptive Navigation**: preserve the merchant's native menu, estimate affect only from limited non-biometric signals, adapt only the app-owned module when confidence is calibrated and high, and use an optional "how is your search going?" self-report as a direct correction signal. Do not claim accurate emotion detection or guaranteed sales lift. Avoid facial/voice recognition and persistent profiling. Shopify's API terms, last updated 7 October 2026, restrict training AI/ML on Shopify-derived or merchant data without the relevant merchant's consent [S8].

Recommendation: keep the concept and name, but sell it as an **emotion-aware, interaction-adaptive navigation aid**, not a mind reader. Start with a consented, transparent affect hypothesis model, high confidence thresholds, a static-menu fallback, and no high-pressure adaptation. Validate classification against voluntary self-reports separately from sales outcomes. Merchant demand, real-store accuracy, and incremental conversion value remain unproven. Treat EU/UK profiling and cookie rules, transfers from Karachi, and Shopify's data-use terms as launch dependencies.

Do this first — prioritized

1. **Keep the product promise truthful:** Emotional Adaptive Navigation estimates likely interaction affect; it does not read minds or diagnose mental health.
2. **Interview 15–20 merchants** in one catalog-heavy segment and test whether state-aware navigation solves a paid problem beyond their existing recommendations.
3. **Run a consented research-panel prototype** with non-biometric signals and an optional "how is your search going?" control; measure participant-held-out state calibration before storefront deployment.
4. **Get EU/UK, U.S., Pakistan, and Shopify/API counsel** to review the exact signal collection, consent flow, transfer map, and model-data terms.
5. **Build the MVP as an affect-aware module, not a theme takeover:** confidence-gated probabilistic rules/model, visible controls, preserved native menu, and static fallback. No camera or microphone.
6. **Test three arms:** static baseline, behavior-only personalization, and Emotional Adaptive Navigation. Analyze affect-classification accuracy separately from conversion lift.

Emotional Adaptive Navigation: concept and signal selection

Working definition

Emotional Adaptive Navigation (EAN) is a Shopify storefront feature that uses non-biometric interaction signals to estimate a shopper's **probabilistic, transient affective state**—for example, likely frustration, confusion, uncertainty, engagement, or confidence—and adapts navigation emphasis, help prompts, layout density, and product suggestions. Each state is a hypothesis with an uncertainty/confidence score, not a diagnosis or a claim to know the shopper's inner state. Interaction friction is a measurable pattern, not itself an emotion. The estimate is session-scoped, does not create a mental-health profile, and never changes price, eligibility, urgency, or access.

The app preserves the merchant's native menu. It adapts only its own optional navigation/recommendation module, with a static menu fallback, visible explanation/control, and reset/disable action. "Interaction friction" is an observable signal; **frustration** is a tentative affective interpretation that requires independent validation. [INFERENCE]

Signal comparison and evidence limits

The studies below report incompatible labels, tasks, metrics, and validation designs. Their figures are **study-specific observations, not a transferable "expected Shopify accuracy."** When a full-text sample or metric was not accessible, that is stated rather than guessed. [OUTSIDE: S33–S44]

Signal	What peer-reviewed evidence supports (setting / sample / result)	Expected accuracy; claims justified / not justified	Difficulty, privacy risk, and EAN recommendation
Mouse/cursor trajectory	Yamauchi & Xiao (2018) extracted 16–26 trajectory features in a fixed choice-reaching task after music/film/image affect induction; 10-fold CV models predicted about **10–20% of variance** in positive-affect and attentiveness self-ratings. The accessible abstract does not report N; it is a lab reaching task, not storefront browsing. Freihaut & Göriz (2021) tested **N=53** within participants across four high-/low-stress lab tasks; self-report/physiology confirmed manipulation, but mouse-use patterns showed **no clear stress differences**. [S33–S34]	The 10–20% is variance explained, **not classification accuracy**. The null stress result is important. No generalizable emotion-classification accuracy can be promised; mouse motion may be a weak supplementary signal for friction/arousal only.	Medium–high; raw pointer paths are granular behavioral monitoring, vary by device and motor/accessibility factors. **Do not record trajectories in MVP.** If later tested, use coarse, short-lived aggregates and explicit consent.
Touch/swipe behavior	Ghosh et al. (2019) evaluated TouchSense in a **3-week in-the-wild Android study with 22 participants**. Personalized models used typing/swiping features plus periodic self-reports for happy/sad/stressed/relaxed labels; mean **AUC-ROC .73 (SD .06), maximum .87**. This is the strongest directly relevant field signal found, but it depends on per-user modeling and a smartphone keyboard, not Shopify navigation. [S35]	AUC is not accuracy and a personalized N=22 result does not imply a new visitor's emotion can be classified at .73. **Justified:** touch patterns may contain affective information in that narrow setting. **Not justified:** store-wide/general-population emotion recognition.	Medium; touch dynamics may reveal typing style, disability, language and identity. **Defer.** Do not collect swipe paths or text content without a reviewed consent design.

Signal	What peer-reviewed evidence supports (setting / sample / result)	Expected accuracy; claims justified / not justified	Difficulty, privacy risk, and EAN recommendation
Keystroke/typing timing	Lee et al. (2014) induced affect with 60 IAPS images and a fixed number sequence in **27 lab participants**; keystroke timing and typing accuracy differed statistically, but emotional effects were small relative to individual variability. Lee et al. (2015) recruited **52**; the main repeated-measures ANOVAs excluded **11 participants** with more than three empty valence/arousal cells (the authors also report a mean-imputed analysis). Incomplete SAM ratings removed 117 of 3,276 trials. The task used 63 auditory stimuli and a fixed 9-digit entry. Arousal accounted for 9.14% of duration variance and 11.48% of latency variance after removing individual differences; typing accuracy did **not** differ significantly. Kołakowska & Landowska (2021): **50 recruited / 49 analyzed**; positive/negative opinion F1 **.555 absolute / .627 relative**; pleasure weighted F1 **.76**, but minority low-pleasure F1 **.20** under class imbalance. Random 10-fold CV was on samples, not participant-held-out. [S36–S38]	These are narrow task-specific associations/classifiers, not general shopper affect accuracy. Epp et al. (2011) collected field keystrokes plus experience-sampled self-reports; Yang & Qin's review reports about **75% average accuracy** across 15 two-class state models, while the accessible abstract describes selected results around **77–88%** and 84% for anger/excitement. The accessible abstract does not state N or a held-out-person validation design; these metrics are not transferable to EAN. [S41–S42]	Very high privacy/accessibility cost: text may reveal sensitive content and timing can fingerprint users. **Do not capture raw keystrokes** or instrument checkout/payment fields. Research only with fixed, voluntary tasks and ethics/legal review.
Scroll depth / upward scrolling	Thomas's web-navigation experiments identify time and upward-scroll amount as predictors of **navigation difficulty**; the accessible abstract provides no participant count or accuracy metric. [S40]	**No affect accuracy established.** It may be a friction feature, not evidence that someone is frustrated, confused, or engaged.	Easy to collect but potentially profiling if persisted. If tested, use coarse session bins only and static fallback; do not use as a standalone trigger.
Dwell / time on page	The navigation literature above relates time-on-page to difficulty; the supplied eye-tracking preprint says long fixations may mean importance or complexity [P3, §2.3, S40]. No study located validates dwell time as a commerce emotion classifier.	**Accuracy unknown.** Dwell can mean interest, reading, distraction, a paused tab, or confusion. No claim of attention, confidence, or frustration is justified from dwell alone.	Low implementation effort; moderate profiling risk when linked across events. Omit from the first model or use only as low-weight corroboration in a consented test.
Search reformulation	No peer-reviewed e-commerce study with validated affect labels and transferable emotion accuracy for query reformulation was verified in this review. [INFERENCE; search not exhaustive.]	**No validated affect accuracy found.** A query change may reflect refinement, a changed goal, vocabulary mismatch, or poor results; it cannot alone establish frustration or uncertainty.	Search text can expose sensitive interests. Keep only a short-lived count/relation between searches, not raw queries; process only with required notice/consent.
Navigation patterns / backtracking	Thomas (2014) found interaction and page-view patterns useful for locating navigation difficulty, not for inferring emotion. Rahman et al. (2024) report sensor-free frustration detectors validated on two independent online-learning datasets, but the accessible abstract does not state N or exact performance metrics. Neither study is retail storefront validation. [S40, S43]	No transferable Shopify affect accuracy is established. The education result supports feasibility of testing interaction-based frustration hypotheses; it does not prove a commerce model will generalize. Treat loops/backtracking as possible friction, not a confirmed feeling.	Medium; event counts are feasible in a theme extension, but histories can become profiles. Use short session windows, require applicable consent, and abstain if the signal conflicts with other evidence.
Cart hesitation / abandonment	No direct peer-reviewed evidence was verified that cart dwell, removal, or hesitation accurately identifies a shopper's emotion. These events have obvious competing explanations—price, shipping, inventory, comparison, payment, interruption. [INFERENCE; no direct accuracy source found.]	**Unknown / do not assign an accuracy.** It may correlate with purchase uncertainty or friction, but it cannot identify emotional vulnerability. Never use it to trigger pressure.	High if linked to order/customer data. **Exclude from affect inference MVP.** Later, test transparent assistance (shipping/returns/comparison), not emotion-targeted offers.

Signal	What peer-reviewed evidence supports (setting / sample / result)	Expected accuracy; claims justified / not justified	Difficulty, privacy risk, and EAN recommendation
Typed search/chat text or sentiment	Kelly & Sharot (2024/2025) analyzed webpage text valence and self-reported mood/mental health; they did **not** infer mood from typed search queries. Query sentiment itself is not a validated proxy for the shopper's internal state in commerce. [S39]	No usable accuracy estimate for storefront query-to-emotion inference was verified. Use queries for stated product intent, not for a hidden affect estimate; do not infer health, religion, financial distress, or mental state.	High if raw queries/chat are retained or sent to a model. For MVP, process only what is needed for search help, discard raw text, and do not call third-party LLMs with it.
Context (device, time, referrer, returning status, cart value)	These are useful context/merchandising variables but are not direct affect measurements; no appropriate emotion-classification accuracy was identified. [INFERENCE]	**No emotion accuracy claim justified.** Device viewport is valid for layout; time/referrer/cart value are weak proxies and can encode sensitive or socioeconomic characteristics.	Low to medium. Minimize and do not use them to label states, target vulnerability, or differentiate prices.
Optional shopper self-report	Kelly & Sharot's four studies (**total N=1,145**) measured mood alongside webpage-text valence; the two observational studies found small associations with psychopathology (**r=.127 and .138**). A randomized browsing study (**N=102**) changed reported mood; an affective-label intervention reduced negative-page selection (**partial η^2=.041**). This is content-valence browsing research, **not e-commerce emotion classification**. [S39]	A direct answer is a criterion for the shopper's *reported* current experience, but it is subjective and not objective ground truth. No classifier accuracy applies. Optional "How's your search going? — stuck / unsure / exploring / doing fine / skip" is user-controlled.	Low–medium if stored; avoid retaining individual responses. Use in a consented validation panel and, in production, keep the answer session-only and correctable.
Camera/facial emotion recognition	Barrett et al.'s 2019 review across cultures/contexts found facial configurations are not sufficiently reliable, specific, or generalizable to diagnose discrete emotions. [S26]	No defensible universal accuracy for emotional diagnosis. A model confidence score is not evidence of validity.	High biometric/image/AI Act and reputational risk. **Avoid.**
Voice emotion recognition	El Ayadi et al. (2011) survey acoustic/prosodic features, classifiers, and speech-emotion databases; it is not a Shopify benchmark or evidence of general field accuracy. No directly applicable storefront validation was verified. [S44]	**Unknown for this use case.** A survey of methods does not justify a universal voice-emotion accuracy claim.	High and unnecessary. Microphone audio can capture speech/content; voiceprints used for unique identification trigger additional biometric risk. **Avoid microphone/voice collection.**

Product position — [INFERENCE]: keep Emotional Adaptive Navigation as the product. A defensible first build can estimate a small set of **probabilistic affect hypotheses** from low-intrusion, session-only signals, but it must communicate that the inference is uncertain and validate it on held-out people. The literature supports a testable hypothesis, not a promise of general affect accuracy or a sales lift.

1. Evidence review: Emotional Adaptive Navigation

Supplied papers: claims, methods, effects, and limitations

Paper	Claim and method	Sample / effect size	Limitations and relevance
P1. "Fractal Adaptive Menus: Concept, Engineering, and Use Case" (Sahraoui, Vanderdonckt & Kieffer, EICS Companion 2025; 8 pages; DOI 10.1145/3731406.3734978)	Proposes a recursive, self-similar hierarchical menu that adapts to screen space and supports frequent navigation through deep structures. Describes a Java prototype and a steel-mill checklist use case. The 30-check workflow is mapped into a menu. [P1, pp. 1–6]	**No empirical participant sample, task-time estimate, error rate, satisfaction score, or effect size is reported.** The real-world case is a prototype/use-case description, not a controlled field evaluation.	The authors explicitly say empirical demonstration remains future work [P1, p. 7]. The mechanism is structural/resolution adaptation, not emotion inference or product recommendation. It supports considering stable, hierarchical navigation patterns—not the core premise. Small-screen submenus may become hard to select [P1, p. 5].

Paper	Claim and method	Sample / effect size	Limitations and relevance
P2. "Integrating Human Feedback into a Reinforcement Learning-Based Framework for Adaptive User Interfaces" (Gaspar-Figueiredo et al., EASE 2025; preprint arXiv:2504.20782)	Adds per-participant preference models to a reinforcement-learning UI framework. Participants compared 64 pre-generated UI-adaptation clips, then used adaptive and fixed interfaces in two lab tasks (e-learning and trip planning); satisfaction and engagement were measured with QUIS and UES. [P2, §§3–5]	**33 convenience-sampled master's CS students** (23 men, 7 women, 3 nonbinary); two-domain within-subject crossover; 66 observations per outcome. Relative to adaptive, non-adaptive scores were lower by **0.97 points on QUIS (1–10; 95% CI 0.74–1.20, p<.001)** and **0.23 on UES (1–5; 95% CI 0.04–0.41, p=.016)**. Means: satisfaction 6.90 vs 5.93; engagement 3.41 vs 3.18. The paper does **not** report a standardized effect size (e.g., Cohen's *d*). [P2, Tables 2–4]	It tests **personalized stated preferences**, not inferred emotional state. It measures UX questionnaires, not conversion, AOV, navigation success, merchant outcomes, or Shopify deployments. Tasks are artificial, sample is small and narrow, participants saw researcher-created clips, and there is no reported a priori power calculation. The framework's training used a separate 25-user e-commerce dataset and a 4090 GPU/1 million RL steps per agent—far beyond an MVP. Results support user control/preference feedback as a hypothesis worth testing, not emotion-based sales lift.
P3. "Eyes Bring out Effectiveness? Applying Eye-Tracking to Explore Consumers' Visual Perception of Horizontal and Vertical Menu Formats in E-commerce Website Design" (provided copy identifies itself as an unreviewed SSRN preprint, SSRN 4920862; author/date not visible in the supplied copy)	Two phases: 100 valid cross-border e-commerce questionnaires plus four expert interviews; then an eye-tracking prototype comparison for a known-item robot-vacuum search. It reports a horizontal main menu with a simplified category layout and a vertical local menu as a proposed improvement. [P3, §§3–6]	Phase 1: 100 valid questionnaires (the write-up discusses US, UK, Japan and Taiwan; its sampling description also mentions China, without matching results). Phase 2: 40 recruited, **39 analyzed** (17 male, 22 female). The paper reports main-menu fixation duration **8.75 vs 4.82** (t=2.421, df=76, p=.018) and "effective focus" **0.90 vs 0.36** (t=8.078, df=76, p<.001); local-menu comparison reports t(76)=−2.082, p=.041. It does not provide a clear satisfaction effect or a reliable, consistently interpretable effect-size estimate. [P3, Tables 3–7]	**Not peer reviewed.** The methods call the experiment within-subject, but the reported tests are independent-samples t-tests with df=76 and describe 39 participants in each version; this is inconsistent with a 39-person within-subject sample. Metric definitions, direction of "effective focus," and the prose interpretation conflict; the satisfaction claim is not backed by a clearly reported between-condition effect. A narrow lab task, one prototype, known-item search, and no real transaction. Treat numbers as exploratory, not a dependable effect. No emotion measure.

What the evidence supports—and does not

- **[PAPER]** P1 suggests a design pattern for hierarchical navigation. It does not show that adaptive menus improve outcomes, and its authors say that remains to be tested.
- **[PAPER]** P2 offers preliminary evidence that **user preference feedback** can improve perceived experience over a fixed interface in two lab domains. It does not compare preference-based adaptation with ordinary clickstream personalization and does not estimate commerce lift.
- **[PAPER]** P3 tentatively points to information architecture, menu complexity, and vertical local menus as factors affecting search performance. Its unreviewed status and method/statistical inconsistencies make the reported significance weak evidence. It says nothing about inferred emotional states.
- **[OUTSIDE]** Barrett et al.'s systematic review of facial emotion expression/perception literature (2019, 20(1), 1–68) reviewed studies across adults, children, cultures, and contexts. It found emotion-category facial patterns can occur more than chance, but are not sufficiently reliable, specific, or generalizable to diagnose a particular emotional state. It is a review, not a new shopping experiment or a single-population effect estimate. [S26]
- **[PAPER]** None of the three supplied papers measures an inferred affective state or tests emotion inference. They therefore **neither support nor refute** the EAN affect-inference component.
- **[OUTSIDE]** Peer-reviewed studies show that non-biometric signals sometimes contain affective information in constrained tasks, but results are mixed, often small, individualized, and not validated on retail navigation. Examples and limits follow.
- **[INFERENCE]** Emotional Adaptive Navigation is therefore **unproven and testable**. The evidence warrants a controlled feasibility study with self-report labels, not abandoning the concept or marketing a proven detector. Conversion uplift and state-classification validity are separate hypotheses.

Additional peer-reviewed behavioral-affect evidence

Study	Design / sample	Reported result	What it does and does not establish
Yamauchi & Xiao, “Reading Emotion From Mouse Cursor Motions” (2018) [S33]	Lab choice-reaching task; 16–26 cursor trajectory features; positive/negative affect induced with music, clips, or pictures and rated by questionnaire. **N was not reported in the accessible abstract.**	10-fold CV models explained about **10–20% of variance** in positive-affect and attentiveness ratings. This is variance explained, not classification accuracy.	Evidence of an association in a tightly controlled reaching task. No validated, general-purpose mood classifier; no commerce/site navigation result. Sample-size and split details require full-text verification.
Freihaut & Göriz, “Using the Computer Mouse for Stress Measurement” (2021) [S34]	Within-participant laboratory experiment, **N=53**, four tasks, high- vs low-stress condition; stress manipulation checked with self-report and physiological data.	Despite successful stress induction, authors found **no clear mouse-usage differences** by stress condition.	A direct null result against assuming mouse patterns reliably reveal stress, even when stress is experimentally manipulated. Stress is not identical to frustration, but it is a close adjacent affective construct.
Epp et al., “Identifying Emotional States Using Keystroke Dynamics” (2011) [S42; summary also in S41]	CHI field study; self-reported states and keyboard rhythm. **N is not given in the accessible abstract/review summary.** The abstract describes 15 affective labels and experience sampling.	The review reports about **75% average accuracy** for two-class state models; the accessible abstract reports selected model accuracies of roughly **77–88%** and 84% for anger/excitement. The validation split and participant-held-out performance were not available in the accessible source material.	Promising, but a high task-specific binary accuracy is not a general emotional-state accuracy. Missing accessible N/split details and the difference between one-vs-rest classification and retail browsing limit transfer.
Lee et al., keyboard typing after visual affect induction (2014) [S37]	**N=27** university participants; 60 IAPS images; fixed number sequence on a numeric keypad; emotional conditions studied with IAPS and self-report.	Key duration/latency/typing accuracy differed statistically, but authors report emotional effects were small relative to individual differences; no emotion-classifier accuracy was tested.	Shows motor behavior can vary with induced affect in a constrained lab task; it does not show that natural shopping keystrokes identify a shopper’s state.
Lee et al., keyboard typing after auditory affect induction (2015) [S38]	**52 participants**; 63 sound stimuli and fixed 9-digit typing task. Eleven participants with >3 empty valence/arousal cells were excluded from the main repeated-measures ANOVAs (**41 analyzed there**); an imputed-data analysis was also reported. Separately, 117/3,276 trials lacking SAM ratings were removed.	Arousal accounted for **9.14%** of key-duration variance and **11.48%** of key-latency variance after removing individual differences; typing accuracy did **not** significantly differ.	Small lab effects in a fixed motor task; exclusions concern affect-label coverage, and results are not a classifier-accuracy estimate or a natural-shopping validation.
Kołakowska & Landowska, keystroke dynamics while writing positive/negative opinions (2021) [S36]	**50 students recruited / 49 analyzed**; lab, four short recalled learning/teacher opinions per person; 51 features; self-reported pleasure/arousal; 10-fold CV on 196 text samples.	Positive-vs-negative **opinion-label F1=.555 absolute / .627 after personal-baseline normalization**. For high/low pleasure, weighted F1 reached .765 under one threshold, but minority low-pleasure F1 was only .203 and classes were imbalanced.	Opinion polarity is not the same as momentary affect. The paper does not report participant-held-out CV; because each person supplies multiple samples, random-fold leakage of individual style is a concern. F1 is not accuracy and does not imply a storefront-ready model.
Ghosh et al., TouchSense emotion detection during smartphone text entry (2019) [S35]	**N=22**, three-week in-the-wild Android study; touch typing and swiping; periodic self-reports; personalized models for happy/sad/stressed/relaxed.	Mean **AUC-ROC=.73 (SD=.06), max=.87** from the combined touch modalities.	Most directly field-like result for a non-biometric interaction signal, but very small and smartphone-specific; personalization and self-report were built into the design. It does not validate new-shopper or cross-store accuracy.
Thomas, “Using Interaction Data to Explain Difficulty Navigating Online” (2014) [S40]	Series of web-navigation experiments using site interaction logs. Exact sample counts/performance figures were not in the accessible abstract.	Time taken and upward scrolling were identified as predictors of **navigation difficulty**.	Supports using behavior to detect possible friction, not to classify frustration, confusion, or another internal emotion. No affect labels or emotion accuracy.
Rahman et al., “Generalisable Sensor-Free Frustration Detection in Online Learning Environments” (2024) [S43]	Peer-reviewed online-learning work; abstract says detectors were validated on two independent datasets from different institutions. The accessible abstract does **not** report N, split details, or numerical accuracy.	The abstract reports cost-sensitive Bayesian classifiers outperformed alternatives with a small generalizable feature set, but provides no metric to quote here.	Evidence that interaction-based frustration prediction is testable in an educational system. It does not establish Shopify accuracy; obtain and audit the full paper before relying on exact metrics or deploying outside education.

Study	Design / sample	Reported result	What it does and does not establish
Kelly & Sharot, web-browsing patterns and mood/mental health (2024 online / 2025 issue) [S39]	Four studies, **total N=1,145**. Studies 1–2 linked browser history and webpage-text valence to self-reported mood/psychopathology; Study 3 randomized negative vs neutral content (**N=102**); Study 4 tested affective labels on search results.	Negative-page valence had small associations with psychopathology (**r=.127/.138**) in two samples; randomized negative content changed post-browsing mood (partial η^2=.151). The labeling intervention slightly reduced negative-page selection (partial η^2=.041).	The signal was **semantic valence of page content**, not mouse/click/dwell or shopping behavior. This is evidence mood and browsing can interact, not a validated classifier. It also warns that adapting content to a low mood can create a feedback loop or affect welfare.

The 2021 KMT review [S41] surveys keystroke, mouse, and touch studies, but is not a new participant study or pooled validation benchmark. Across the primary studies above, headline metrics are study-specific and generally depend on fixed tasks, induced states, self-report labels, or personal models. The defensible target for an MVP is not a universal "emotion accuracy" number: it is a calibrated probability of a narrow, shopper-relevant state in one supported storefront context, with abstention when uncertain.

Conclusion: Emotional Adaptive Navigation is **unproven and testable**. The supplied papers neither support nor refute its inference layer; outside behavioral-affect studies provide a reason to test it carefully, not to claim established accuracy or sales benefit.

2. Emotional Adaptive Navigation: novelty and market validation

Competitive position

The feature space is already crowded for adjacent jobs:

Product / category	What it already does	Implication
Shopify **Search & Discovery** (free)	Search tuning, filters, synonyms, product boosts, recommendations, and discovery analytics. [S27]	A generic "better recommendations" pitch competes with a built-in free baseline.
Nosto; Algolia; LimeSpot; Rebuy	Search, merchandising, recommendation widgets, category ranking, cross-sell/upsell, personalization, and analytics. [S28–S29]	Large vendors already sell behavior-driven discovery. Competing on "AI recommendations" alone is weak.
Meteor Mega Menu and other navigation apps	Theme-based menus and menu presentation. [S30]	A menu builder is also not new; theme compatibility and merchant workflow matter.
Skinalyze and Beautyxia (adjacent, not direct)	Shopper-submitted face/selfie analysis for skin attributes and product matching. App Store listings show Skinalyze from $9.99/month (one review) and Beautyxia free/beta (one review). [S30]	Facial analysis exists in a narrow beauty use case, but is not evidence of defensible emotion recognition or general demand. It increases privacy/regulatory sensitivity.

Gap assessment — [OUTSIDE + INFERENCE]: In the App Store pages and search results reviewed on 21 Oct 2026, I found abundant search/recommendation/personalization and some shopper-submitted skin-analysis tools, but no clearly established Shopify product whose core proposition is **Emotional Adaptive Navigation**. This is **not proof no such tool exists**; App Store search results were incomplete and products change. The potential wedge is an emotion-aware navigation layer that is transparent about uncertainty, preserves native menus, and adapts only a merchant-approved extension. Absence of a direct competitor does not prove demand or defensibility.

Target segments and willingness to pay

Start with one niche where navigation is genuinely hard: apparel with many collections/variants, beauty with routine/concern navigation, home goods with a broad catalog, or specialty retail with technical attributes. The buyer is likely a Shopify store owner or e-commerce lead already spending on conversion, search, or merchandising. Prefer merchants able to support consented experiments and enough traffic to evaluate outcomes; low-traffic stores may still be research/design partners but cannot reliably measure incremental sales. Beauty/health and child-directed merchants need heightened legal review because affect labels and shopping context can drift into health/vulnerability inferences.

Willingness to pay is **unproven**. Do not use competitor list prices as proof. Ask merchants what problem they pay to solve today, their existing stack, the minimum measurable gain that would justify another app, and whether they would pay for a pilot. Validate on a clickable prototype/concierge before producing a Shopify extension.

Cheap validation plan

1. **15–20 problem interviews** over two weeks; recruit one vertical with complex catalogs. Ask about frustration/uncertainty at discovery, what customers do when stuck, current app spend, and privacy/brand acceptability. Do not lead with a claim that the app can read emotion.
2. **Landing-page and prototype test:** retain the product name. Test truthful lines such as “Emotional Adaptive Navigation: adapts navigation to how shoppers interact” and “emotion-aware suggestions, with shopper control.” Measure qualified merchant demo bookings and paid-pilot commitments, not vague consumer interest.
3. **Consented concierge/research pilot:** 3–5 stores plus a separate shopper panel. Use a prototype “How’s your search going?” control; compare interaction-derived affect hypotheses to repeated voluntary self-report. Do not record raw typing or pointer paths.
4. **Go/no-go:** continue only if merchants pay/commit, the consent model is workable, and participant-held-out affect prediction performs better than a prevalence-only baseline with acceptable calibration/false positives. If sales outcomes are underpowered, say so; do not change the product name or claim success from click-through alone.

3. Emotional Adaptive Navigation: legal and compliance — primary launch risk

This section describes likely issues, not a legal conclusion for a particular merchant or country. The app can have merchants in the EU/UK/California while its developer is in Karachi. GDPR/UK GDPR applicability is based on the data subjects, offer/monitoring, and processing—not just where the developer lives. Ask counsel to review the final data flow and customer notice before beta.

GDPR / UK GDPR, profiling, and lawful basis

EU GDPR. A store visitor’s page views, device identifiers, search queries, click paths, and inferred affect scores can be personal data when they relate to an identifiable or singled-out visitor. Systematically predicting a person’s mood, confidence, interests, or behavior from those data is likely **profiling** within Article 4(4), even if the output is only a probability and the inference is non-biometric. GDPR may apply to a Pakistan-based developer under Article 3(2) when it offers services to or monitors people in the EU. Profiling is not categorically prohibited, but requires a lawful basis, purpose limitation, minimization, transparency, security, rights handling, and potentially a DPIA. An Article 6(1)(f) legitimate-interest basis may be arguable for low-impact first-party EAN after a documented necessity/balancing test, clear notice, and objection path; it is not a blanket permission for invisible tracking, cross-site profiling, or sensitive emotion inference. Consent is often the more defensible path for optional behavioral tracking. [S1, Arts. 3–6, 13–14, 21, 25, 35]

An inferred “friction” or “uncertainty” score is **not automatically** special-category data. The risk changes if the system deliberately infers or uses mental-health status, disability, health conditions, or another Article 9 category, or if the context makes an affect score reveal health information. Keep state labels about the shopping interaction, not the person’s psychological condition; don’t infer “anxiety disorder,” “depression,” vulnerability, or capacity. [S1, Arts. 4(15), 9]

Article 22 applies to decisions based **solely** on automated processing, including profiling, that have legal or similarly significant effects. Ordinary product ranking or showing a different category is unlikely by itself to meet that threshold, but price/credit eligibility, access restrictions, sensitive categories, children, or manipulative profiling can change the analysis. Article 22 does not erase the other profiling/transparency/lawful-basis duties. Where special-category data is involved, the Article 22 exception is narrower. [S1, Arts. 9, 22; Recital 71]

UK divergence as of this report. The Data (Use and Access) Act 2025 amended UK GDPR automated-decision provisions; the changes to Article 22 took effect **5 February 2026**. The UK now defines “solely automated” and “significant decision” in Articles 22A–22D. Significant decisions based partly on special-category data remain restricted (with specified exceptions); for other significant solely automated decisions, safeguards must provide information, representations, human intervention, and contestability. Decisions

cannot rely on the new “recognised legitimate interest” basis. The ICO’s detailed post-Act guidance was updated 31 March 2026 but was still under consultation. Ordinary product recommendations are generally less likely to be significant; the ICO warns that context, children, behavior, and discriminatory or consequential offers can make a nudge significant. [S2–S3]

Consent versus legitimate interests: these are not interchangeable. The merchant normally determines why/how the storefront personalization works; the app may be its processor for those operations. If the developer uses customer events for its own analytics, cross-merchant benchmarking, model improvement, advertising, or another purpose, it may become an independent controller for that processing and needs its own basis/notices/rights process. Avoid that secondary use by default. For direct marketing, respect the absolute right to object to processing for direct marketing, including related profiling. Under the UK Act, the ICO says the legitimate-interests basis may be possible for some ADM, but requires a purpose/necessity/balancing assessment; UK “recognised legitimate interests” is not available for ADM. [S2–S3]

Representatives and officers: a non-EU developer subject to GDPR under Article 3(2) will generally need an EU representative under Article 27 unless a narrow exception applies; a similar UK representative question arises under UK GDPR. A DPO is not automatically required for a solo app, but may be mandatory if core activities involve large-scale, regular/systematic monitoring or large-scale special-category processing. Counsel should assess this from actual volume and purposes. [S1, Arts. 27, 37]

Cookies, pixels, and ePrivacy / PECR

EU ePrivacy Directive Article 5(3) generally requires prior consent for storing information on or accessing information from a user’s terminal equipment, except for transmission and strictly necessary service functionality. This is separate from the GDPR lawful basis: legitimate interests does **not** override a cookie-consent requirement. Cookies are not the only covered mechanism; local/session storage, fingerprinting, pixels, or other access to device data can be in scope. [S4]

UK change: from 5 February 2026, DUAA amended PECR Regulation 6 and added Schedule A1. Alongside consent and strictly necessary exceptions, there are narrow exceptions for some statistical collection (with clear information and a free simple opt-out; it does not cover automatically emitted device data) and website appearance/functionality adaptation (clear information and simple opt-out). This might cover a user-selected UI preference in some implementations; it is **not a safe assumption** for emotion inference, granular behavioral logging, or a Shopify pixel that profiles automatically emitted events. Obtain a UK PECR review. [S5–S6]

Shopify’s current guidance says storefront `shopify:*` events fire regardless of buyer tracking consent; using them directly does not remove the developer’s consent obligations. Shopify recommends consent-aware Web Pixels for analytics/marketing, with declared purposes (`analytics`, `marketing`, `preferences`, `sale_of_data`) and its Customer Privacy API. Set only the purposes actually needed. Pixel sandboxing is not a legal exemption. [S7–S8]

California CCPA/CPRA

The CCPA/CPRA generally covers a for-profit business doing business in California that meets a statutory threshold: annual gross revenue above **$26.625 million** (2025-adjusted), buying/selling/sharing personal information of at least 100,000 consumers/households, or deriving at least 50% of annual revenue from selling/sharing personal information. Browsing history, device identifiers, and inferences—including an affect score associated with a visitor—can be personal information. Service providers/contractors have contract-based restrictions when processing for covered businesses; using the score for the app developer’s own cross-context advertising or other unrelated purpose can change that role. The merchant will often be the business; the app may be a service provider if the written contract restricts use to specified business purposes and bars selling/sharing or unrelated use. If the app independently determines purposes, it may incur separate obligations. Support access, correction, deletion, opt-out of sale/sharing (including cross-context behavioral advertising), Global Privacy Control handling where applicable, and sensitive-personal-information restrictions as contractually assigned. Do not assume a small app is automatically outside every obligation: its merchant relationships, own business, and service-provider status matter. [S9, S31]

California’s 2025 regulations became effective **1 January 2026**, but CPPA says the ADMT significant-decision requirements begin **1 January 2027**. These apply to defined significant decisions, not automatically every product recommendation. A normal storefront recommender is unlikely to be a “significant decision”; consequential eligibility, financial, health, or similarly significant personalization needs specialist review. [S9–S10]

Biometrics, facial analysis, and the EU AI Act

Under GDPR, biometric data is data from specific technical processing of physical, physiological, or behavioral characteristics that allows or confirms unique identification; it is special-category data when processed **for uniquely identifying** a person. A photo is not automatically Article 9 biometric data merely because it depicts a face, but facial geometry templates, identification, and emotion/health inference materially increase the risk. Inferred health/mental-health data can also raise Article 9 concerns depending on what is inferred and how it is used. [S1, Arts. 4(14), 9]

Illinois BIPA is a major U.S. litigation risk: it covers “scan of face geometry” and voiceprints as biometric identifiers, with written notice/release, publicly available retention/destruction policy, and restrictions on disclosure and profit. It has a private right of action. A storefront selfie/emotion feature could implicate BIPA for Illinois residents even if the business is based elsewhere. Other state biometric/privacy laws may also apply. **Do not build this for the proposed MVP.** [S11]

The EU AI Act defines an “emotion recognition system” as inferring emotion or intention **based on biometric data**. An EAN model using only clicks, search reformulation, or other non-biometric behavior is therefore outside that specific definition on the current text; it is **not** outside GDPR profiling, ePrivacy, consumer-protection, or Shopify rules. If camera/voice/biometric inputs are added, Article 5(1)(f) prohibits emotion inference in workplaces and educational institutions (with medical/safety exceptions), and Article 50(3) requires deployers of permitted emotion-recognition/biometric-categorisation systems to inform exposed people at first exposure and process personal data under applicable privacy law. Article 5(1)(a) prohibits certain manipulative/deceptive AI techniques that materially distort behavior and cause or are likely to cause significant harm; Article 5(1)(b) addresses exploitation of age, disability, or specific social/economic vulnerability causing significant harm. This is not a ban on every product recommendation, but EAN must not exploit a state estimate to impair choice or target a vulnerable shopper. The current Commission Service Desk text is consolidated as of 27 July 2026. [S12–S13]

Pakistan position and international transfers

Pakistan: the official Senate record for the 2023 private member Personal Data Protection Bill says it was introduced on 13 February 2023 and later “stands withdrawn from the Committee”; a MoITT-hosted May 2023 draft remains a draft. The official materials reviewed do not establish an enacted, comprehensive national data-protection statute as of this report. Pakistan’s Constitution Article 14 protects dignity and privacy of the home, but is not a comprehensive data-protection statute; sector/cybercrime/electronic-transaction rules still matter. Do not market “Pakistan PDPA compliance” as a substitute for review. Legislative status could have changed after the accessible record, so verify the Gazette and obtain local counsel before launch. [S14–S15, S32]

For EU visitor data accessed or stored from Karachi, Pakistan is not identified as an EU adequacy jurisdiction in the sources reviewed. If this is a restricted transfer, the EU exporter may need an appropriate Article 46 mechanism (often SCCs) and a transfer impact assessment/supplementary measures. The UK has its own transfer regime and may require an IDTA or UK Addendum. Hosting in an EU region does not necessarily eliminate a transfer if Karachi-based staff can access the data. Minimize access and have counsel map every vendor/subprocessor and support path. [S1, Arts. 44–49; S16]

Shopify requirements and platform boundaries

Requirement	Planning consequence
App Store review and truthful listing	The app must actually work as described; request only required scopes; include privacy policy, demo/test credentials and accurate setup steps. Shopify rejects unsupported claims and checkout manipulation. Use GraphQL Admin API for a new public app, not legacy REST. [S17]
Protected customer data	Public-app access to customer/order data needs Shopify approval and minimum-necessary justification. Avoid customer names, emails, orders, and account history in MVP. Shopify’s Level 1 requirements include minimization, purpose limitation, consent decisions, opt-outs, merchant DPA, retention, encryption; Level 2 adds backup encryption, environment separation, access controls/logging, and incident response. [S18]

Requirement	Planning consequence
Mandatory GDPR/privacy webhooks	Every App Store app needs `customers/data_request`, `customers/redact`, and `shop/redact`, even if it claims to collect no personal data. Verify HMAC, acknowledge, fulfill within Shopify's deadline; customer/shop redaction must connect to actual deletion paths. [S19]
Shopify API terms	Use merchant data only to provide EAN; delete originals/copies within 30 days on uninstall/no longer needed/request; provide access/correction; service providers must contractually limit purpose and protect data. Report suspected data breach to Shopify within 24 hours. Shopify's 7 Oct 2026 terms restrict using Shopify-derived/merchant data (including aggregate/derived forms) to train/fine-tune/improve ML/AI, unless Shopify gives written consent or the relevant merchant consents for its store/data. Do not train on visitor affect labels/events or pool across tenants without explicit, reviewed opt-in. [S8]
Theme app extensions	Modify storefront themes only through Theme App Extensions; no direct theme edits or theme-code injection as the core integration. App blocks require merchant placement; app embeds are merchant-activated and can run in vintage and OS 2.0 themes. Extensions cannot render on checkout pages. [S20]
Web Pixels	Pixels run in a strict sandbox, expose selected Shopify events and privacy status, and require `write_pixels` + `read_customer_events` to create/update. They are for consent-aware event collection, not a universal way to change the DOM or reorder theme navigation. [S7–S8]
Checkout	Do not adapt checkout based on inferred states in MVP. Checkout customizations must use documented extensions, cannot add countdown timers or ask for payment data, and optional charges require clear, explicit opt-in; cheapest shipping defaults remain protected. `checkout.liquid`/scripts have been sunset on relevant checkout/order-status surfaces. [S21]
Billing	New public apps should use Shopify App Pricing when supported; external billing is generally not allowed for App Store apps. [S22]

Merchant versus developer roles, notices, DPA, retention

Roles follow actual purposes, not labels in a contract. A likely starting point is: **merchant = controller/business** for storefront personalization; **app developer = processor/service provider** for operating the feature under merchant instructions. The developer is likely an independent controller for its own account administration, billing, fraud/security, and any separately decided product analytics. Joint controllership is possible if the parties jointly decide purposes/means. Have counsel map each activity.

Before beta: publish an app privacy policy (Shopify requires it), merchant terms, and an Article 28/CCPA service-provider DPA covering categories/purposes, instructions, confidentiality, security, subprocessors, international transfers, rights-request assistance, breach notice, deletion/return, audits, and no model-training/secondary use by default. Give merchants a customer-facing notice template describing the actual event tracking, personalization, consent choices, and opt-out. [S8, S18–S19]

Suggested data policy: keep interaction features in browser memory for the active session; do not persist raw clicks, text, pointer movement, or video. If a visitor opts into a test, use a random experiment assignment with a short expiry; retain only merchant/day/variant aggregates after applying a minimum cohort size. Do not include IP, raw query, customer ID, or exact timestamps in analytics. Delete tenant data after uninstall within Shopify's 30-day terms; keep audit records only as long as needed for security/legal obligations and explain any non-deletable record. These are product recommendations, not statutory safe harbors.

Dark-pattern and manipulation risk

The FTC's 2022 *Bringing Dark Patterns to Light* report describes interface tactics that trick or trap consumers (e.g. hidden terms, obstructed cancellation, false urgency, and manipulative choice architecture). U.S. FTC Act §5 can reach deceptive/unfair conduct. The EU UCPD prohibits misleading/aggressive commercial practices; DSA Article 25 prohibits online platforms from designing interfaces that deceive/manipulate or materially impair free/informed decisions (its direct scope depends on the service). [S13]

The reputational line is sharper than the technical line: using a "frustration" estimate to push a higher-margin product, fake scarcity, discount, subscription, or premium checkout is a vulnerable-shopper manipulation story even if no explicit "emotion" label is visible. Keep prices, ranking logic, and major category availability stable; make suggestions visibly optional and explainable; never hide cheaper/relevant items, manufacture urgency, or use inferred stress/financial constraints to increase pressure.

Safe / gray area / avoid

Status	Examples	Conditions / reason
Lower risk / proceed carefully	Optional "How's your search going?" control; session-only low-level interaction features processed in memory; merchant-curated state-specific UI; user can dismiss/reset; static default preserved.	Still profiling/personal-data processing where a visitor is identifiable. Provide clear notice, applicable consent, shopper control, security, access/deletion path, and merchant settings. Lower risk is not a legal safe harbor.
Gray area / counsel + consent design	Probabilistic affect scores from click/touch/navigation traces; confidence-gated menu emphasis; persistent visitor profiles; query sentiment; mood self-report retained beyond session; cross-border analytics or state model; age/health/product-interest segmentation.	Assess GDPR/UK lawful basis and Article 22/UK Articles 22A–D, ePrivacy/PECR and Shopify consent state, CCPA roles/opt-outs, DPIA, transfer mechanisms, retention, accuracy, and Shopify model-training terms. Keep state labels narrow and interaction-specific.
Avoid for this product	Camera/voice emotion recognition; hidden keystroke/session recording; mental-health or vulnerability diagnosis; adapting price/credit/urgency; targeting a negative state to push a sale; suppressing options; cross-store identity; cross-merchant model training without written scoped consent.	Weak evidence and disproportionate biometric/privacy/manipulation risk; potential special-category, AI Act, BIPA, Shopify policy, and reputation exposure.

Where a lawyer must be consulted

Before any live pilot involving behavioral data: (1) EU/UK privacy counsel for Article 3 scope, lawful basis, cookie/pixel consent, DPIA, transfers, and profiling; (2) Pakistan counsel for the developer's local obligations and latest bill/gazette status; (3) U.S. privacy counsel before facial/image features or broad U.S. distribution (BIPA and state laws), and before relying on CCPA service-provider treatment; (4) Shopify/technology counsel to interpret Shopify API Terms §2.3.24 for any model improvement or cross-merchant aggregates. Get a separate consumer-protection review if the app makes urgency, discount, pricing, health, or vulnerability-based changes.

4. Emotional Adaptive Navigation technical architecture

Recommended Shopify fit

Keep the scope small. The embedded Admin app handles onboarding, merchant configuration, consent/purpose controls, experiment setup, and aggregate reports. A Theme App Extension (app block) renders the EAN navigation/recommendation module. A small app embed can collect a tightly allow-listed set of in-session interaction features; a Web Pixel is optional for separately consented evaluation events, not the adaptation engine. Use Shopify's Storefront API `productRecommendations` as one candidate source and merchant-selected collections/tags as another. Admin GraphQL is for merchant-owned catalog/settings; avoid customer/order scopes. [S7, S17–S20, S23]

Do not promise automatic rewriting of native menus. Shopify's Admin API can retrieve menu data, but Theme App Extensions do not safely replace every merchant theme's header behavior. Theme App blocks require placement in supported theme sections; app embeds are independent blocks, not a platform-wide "reorder the header" API. In MVP, keep the merchant's primary menu intact and show a clearly labeled alternate discovery module. If later offering a replacement navigation widget, scope it to selected compatible themes and require merchant preview/approval. [S20, S23]

Data flow sketch

`Shopify Admin (merchant catalog + EAN settings)` → `per-shop config` → `Theme App Extension` → `in-memory session feature extractor + affect estimator` → `state probabilities + confidence` → `merchant-approved EAN module or static fallback` → `optional self-report/correction` → `consented, minimized evaluation events` → `per-store aggregate report`.

Avoid customer/order API scopes, persistent profiles, full session replay, raw keystroke/pointer capture, camera/microphone access, and checkout extensions in MVP. Keep feature extraction and estimation in the browser; server-side services handle merchant config, product candidate sync if needed, subscriptions, privacy webhooks, and consented aggregates. Do not send raw customer data, queries, or inferred state to an LLM/external model provider.

Where inference runs

For MVP, estimate a small state vector: `P(frustration)`, `P(confusion/uncertainty)`, `P(exploratory engagement)`, and `P(goal confidence)`, plus a separate observable interaction-friction score. In early research, a rule-based score is only a provisional score; do not label it a probability or show it to shoppers until it has been calibrated against self-report on held-out participants. After feasibility passes, compare an interpretable calibrated logistic/multinomial model with the rule baseline. A score is a hypothesis, not an emotion truth. Begin with event counts and sequences, not raw trajectories: search reformulations, backtracks, repeated category changes, optional "compare" actions, coarse dwell/scroll bins, and a voluntary "How's your search going?" response.

Provisional confidence gates [INFERENCE; calibrate before launch]: (a) apply only a subtle, app-module change when the top calibrated probability is ≥0.75, exceeds the second-highest state by ≥0.15, and at least two independent signal groups agree; (b) for 0.60–0.75, show only an optional factual help prompt—do not reorder; (c) below 0.60, conflicting signals, missing consent, first visit, or any model/data failure, render the static default. These thresholds are starting policy, not research-established cutoffs; set final thresholds from participant-held-out validation and a false-positive cost analysis. Explicit user selections override model scores.

Keep the affect estimate ephemeral in memory for the current session. If a shopper self-reports, use the response as an explicit current preference and a validation label only under the separately explained research consent. Never join it to a customer account or write it to a Shopify customer metafield. Browser-side inference avoids per-event network latency and reduces data sent to Karachi-hosted infrastructure.

Server-side inference is only justified if a validated use case needs catalog-wide ranking or aggregate merchant reporting. Train any general model using a separately consented research dataset, not Shopify event data. Any future per-merchant updating on Shopify-derived signals requires relevant merchant consent under the API terms, plus the shopper-side lawful basis/notice; no pooled tenant training by default. [S8]

Stack, tenancy, and performance

Recommendation [INFERENCE]: Shopify's current React Router app template with TypeScript and Polaris/App Bridge for the embedded app; Shopify Theme App Extension (Liquid + small TypeScript/JavaScript bundle) for the storefront; Admin GraphQL API for merchant setup; Storefront API for public product data; PostgreSQL for shop-level configuration and webhook state. Start without Redis, a vector database, or an ML service. Use a managed app host and managed Postgres with an EU region available; encrypt secrets/tokens with the provider's managed key facility, use tenant/shop ID in every row and query, and add row-level isolation tests. Keep logs free of visitor IDs/raw query data. Select hosting only after reviewing subprocessors, data residency, incident terms, and a DPA.

The storefront bundle must be asynchronous/non-blocking, cache catalog/config, avoid layout shift, and never delay the merchant's header, product rendering, cart, or checkout. Aim for single-digit KB compressed initial JS as an engineering budget, no long tasks, and client-only adaptation under a few milliseconds. If a network recommendation endpoint is slow/unavailable, render the

static/default widget. Shopify's App Store performance requirement: no more than a 10-point Lighthouse storefront score reduction; test home, product, collection on real themes. [S17, S24]

Metafields/metaobjects are suitable for merchant-owned configuration or product metadata where useful; do not store inferred visitor states in customer metafields. App-owned database settings are simpler for MVP. Admin scopes should be minimal—likely read products and any explicitly justified menu/content access; request Web Pixel scopes only if enabling the analytics feature. Shopify reviews protected data permissions individually. [S18, S25]

5. Emotional Adaptive Navigation: state-to-navigation adaptation and guardrails

Probabilistic affect hypothesis	EAN navigation/product changes	Guardrails
Possible frustration (repeat reformulations + backtracking/error pattern; confidence-gated)	**Menu/order:** keep the native menu fixed; within EAN, emphasize search, filters, "browse all," and factual shipping/returns links. **Help:** "Want help narrowing results?" **Suggestions:** exact/near matches and filter shortcuts. **Density:** compact first view (e.g., 3–4 options) with "see all."	Do not state "you're frustrated"; no discount, urgency, premium upsell, or vulnerability targeting. Shopper can dismiss/reset.
Possible confusion / uncertainty (repeated category changes or option toggles; weak proxy)	**Menu/order:** put plain-language top-level category labels and comparison paths first in the EAN module. **Help:** offer size/spec/returns explanations. **Suggestions:** comparison attributes and concise product alternatives. **Density:** compact, scannable module with visible "see all."	Do not hide alternatives, decide for the shopper, or treat intentional comparison as dysfunction.
Exploratory engagement (broad category/product exploration; low-specificity)	**Menu/order:** emphasize broad, high-level collections before narrow subcategories. **Help:** optional "Explore collections." **Suggestions:** a small, diverse merchant-curated set. **Density:** a few varied tiles, no automatic expansion; full menu/search remain visible.	No endless feed, interruption, auto-scroll, or optimizing for time-on-site. "Engagement" remains a hypothesis, not a shopper label.
Goal confidence / confident known-item search (consistent query/product path; hypothesis)	**Menu/order:** emphasize the matching category, exact-match item, and direct filters. **Help:** show variant/stock detail or a clear route back to search, not a persuasion prompt. **Suggestions:** exact match and materially relevant variants; comparison/support remain available. **Density:** standard density, no forced focus.	Do not suppress lower-priced/relevant options, infer willingness to pay, or equate a consistent path with purchase readiness.
No dominant affect / low confidence / no consent	**Menu/order:** use the merchant's static default. **Help/suggestions/density:** unchanged; optionally show the user-controlled "How's your search going?" control.	No model-driven changes, persistent profile, or degradation for declining consent.

Keep the top-level native menu stable. Within EAN's module, small ordering/emphasis changes are acceptable if the visitor can see all options, understand that Emotional Adaptive Navigation uses current-session interaction patterns, dismiss it, choose a mode manually, reset to default, and use it with keyboard/screen readers. Do not expose an individual named state to merchants or shoppers by default; show an explanation of the signal/action, not a psychological label. Do not alter font size, contrast, motion, or navigation focus in ways that impair accessibility. Respect reduced motion, zoom, touch targets, keyboard navigation, and localization. Merchants preview/approve every experience.

Concretely, only reorder or emphasize links **inside the app-owned EAN module**; never remove a native category. The table specifies the proposed category emphasis/order, help prompt, suggestion type, and density per hypothesis. Any lower-density module preserves a clear "see all" route. Use factual, non-diagnostic prompts ("Want help narrowing results?"), not "You seem frustrated" or "You look ready to buy."

6. Emotional Adaptive Navigation roadmap

Effort assumes one experienced full-time developer; legal/privacy work and merchant recruiting are separate dependencies. These are estimates, not commitments.

Phase	Duration / solo effort	Deliverables	Dependencies and exit criteria
0. Merchant/problem validation	2–3 weeks; 1–2 dev weeks	15–20 merchant interviews; one vertical; EAN naming/claims test; clickable storefront prototype; signal/data-flow map; privacy and Shopify-terms checklist.	Require ≥3 credible paid-pilot commitments and a stated navigation problem not already solved by current tools. Otherwise do not build the full app.
1. Affect feasibility prototype	3–5 weeks; 2–4 dev weeks	Consented adult research panel (suggested 30–50 participants for feasibility, not universal validation); whitelisted session interaction features; optional "How's your search going?" check-ins; operational affect-state labels; self-report and navigation-task measures. Mouse/touch/typing traces, if studied, stay in a fixed research task and are not collected from ordinary storefront users.	Pre-register labels, task scenarios, retention, and analysis. Hold out participants—not events—during validation. Continue to an estimator only if candidate signals beat a prevalence-only baseline on pre-set per-state precision/recall and calibration; otherwise redesign features/thresholds, not the EAN concept. No live inference yet.
2. MVP / private alpha	8–12 dev weeks	Shopify OAuth/admin setup; merchant-controlled EAN block/embed; interpretable affect probabilities from research-approved signals; confidence gate/abstention; static fallback; optional self-report/correction; native menu untouched; consent-aware pixel only if required; privacy notice/DPA; mandatory webhooks; aggregate state/UX metrics.	No customer/order scopes, raw typing or pointer paths, persistent state profile, individual emotion display to merchant, or training on Shopify data. Exit: legal review, held-out validation report, reversible accessible UI, deletion/webhooks tested, 3 theme/device setups pass, Lighthouse delta ≤10 points.
3. Closed beta	4–6 weeks; 4–7 dev weeks	3–5 merchant stores; fixed/behavior-only/EAN experiment assignment; state calibration dashboard with abstention and per-class errors; merchant kill switch, support docs, incident runbook.	Visitor consent and merchant DPA in place; use only stores with eligible traffic. Exit: no critical accuracy, privacy, accessibility, or performance issue; merchants understand EAN is probabilistic; at least 2 of 3 pilot merchants willing to pay/continue. Sales lift is exploratory until adequately powered.
4. Public launch	4–8 weeks; 4–8 dev weeks	Shopify App Store listing and screencast; Supported-theme matrix; Shopify App Pricing; privacy policy, DPA, subprocessor list, setup/help center, monitoring and on-call plan.	Shopify review passed; legal sign-off; privacy rights/deletion flow operational; claims use "emotion-aware / estimates interaction affect," not "accurately detects emotion." Launch only with support capacity and a remote disable switch.
5. Later research / scale	After demand and adequate evidence	More locales/devices; per-merchant calibration; richer state set; consented store-level reporting; possible explicit self-report personalization.	Demonstrate incremental value over behavior-only and held-out-user affect validity. Any per-merchant learning on Shopify-derived events needs merchant consent and a separate legal review. No camera/voice roadmap.

7. Emotional Adaptive Navigation evaluation and proof

Experiments

Use two separate evaluations: **(1) affect inference validity** and **(2) whether EAN changes shopping outcomes**. Do not use a sales lift to claim the state classifier is accurate, or a classifier score to claim the feature improves commerce. Pre-register the primary outcome, state labels, event window, confidence threshold, and stopping rule.

Commerce A/B design (three arms):

- **A — Static control:** merchant's fixed/native navigation and no EAN adaptation (or the existing discovery module).
- **B — Behavior-only personalization baseline:** current-session behavior changes the app module but does not infer or use an affect state (for example, a search rephrase can surface help).
- **C — Emotional Adaptive Navigation:** the same allowed signals feed the confidence-calibrated affect estimator; only high-confidence affect hypotheses modify EAN's app-owned module. Low-confidence sessions abstain to the same static default.

Randomize at visitor/session level only where lawful and with the required consent; otherwise use a user-controlled test cohort or a carefully designed store/time switchback. Keep variant assignment stable within the permitted session, avoid overlapping merchandising experiments, stratify by device/store only using lawful aggregate data, exclude bots, and run through full weekly cycles. Predefine stopping rules; do not repeatedly peek and stop on a temporary lift. Compare **C vs B** for the incremental value of the emotion layer, and **B vs A** for the value of personalization in general.

Affect validation panel: recruit an opt-in panel on a representative demo/merchant storefront. At planned intervals or after a friction event, optionally ask "How's your search going?" with choices such as "stuck," "unsure," "exploring," "confident," "doing fine," and "skip." A separate validated valence/arousal self-report scale can be used in a research protocol if necessary; do not ask for diagnoses or mental-health history. Treat self-report as the best available subjective criterion, not objective ground truth. Collect repeated reports per participant; record device/input method and task; include "skip." Split training/testing **by person** and later by store/device so visits from the same person cannot leak across sets. The feasibility panel is not enough to certify broad population accuracy.

Metrics

Type	Measures
Primary business	Conversion per eligible session; add-to-cart per eligible session; gross margin/contribution where merchant can provide reliable aggregates.
Discovery/diagnostic	Recommendation CTR; search reformulation; time-to-first relevant product; category-to-product path completion; navigation depth; no-result search rate. These diagnose the feature; they are not equivalent to revenue.
UX/guardrails	One-click "helpful?"/satisfaction sample; bounce/exit; return/refund/cancellation; complaints; menu accessibility success; page load/Lighthouse/Core Web Vitals; no unintended hiding/price differences.
Emotion-classification validation (separate from sales)	Per-state precision/recall, macro-F1 and balanced accuracy; one-vs-rest ROC-AUC only with enough data; confusion matrix; Brier score/expected calibration error; abstention rate; false-positive rate at the chosen confidence threshold. Compare predicted affect hypotheses with the shopper's voluntary momentary report. Never call clicks/dwell "ground truth emotion."
Fairness / robustness	Repeat validation by device (touch/desktop), language/locale and accessibility/input mode where panel consent and sample sizes permit. Check class imbalance, per-group false positives, and calibration. Do not collect protected-trait data just to fill a scorecard; use accessible qualitative testing where quantitative subgroup cells are too small.

Power reality: at a 2% baseline conversion, detecting an increase to 2.2% (0.2 percentage point absolute; 10% relative) with conventional 80% power and 5% two-sided significance needs roughly **77,000 eligible sessions per arm** before exclusions/cluster effects. The exact number changes with baseline, randomization, and variance. Most small Shopify stores will not reach this quickly. Report confidence intervals and minimum detectable effect, pool only with explicit merchant permission and a defensible analysis plan, and never call an underpowered test "proven."

The emotion layer's incremental value is **EAN (C) vs behavior-only personalization (B)**, not merely EAN vs static. The self-report panel evaluates whether behavioral signals estimate reported affect; the storefront A/B evaluates whether acting on the estimate changes outcomes. Report null results, per-state errors, calibration, and abstention. Shopify terms restrict training on Shopify-derived merchant data without consent; any use of that data for model improvement or cross-store pooling needs explicit, scoped consent and counsel review. [S8]

8. Emotional Adaptive Navigation: business model and go-to-market

Pricing and costs

Start with a **flat subscription**, not revenue share/attributed sales: attribution disputes and low-volume noise will erode trust. Test (not assume) something like **$29/month starter**, **$79/month growth**, and **$199/month multi-store/advanced reporting**, with a 14-day trial and simple cancellation. Consider a free prototype only if it does not become an unsupported support burden. Price based on features/catalog or transparent session bands; if bands require tracking, explain how sessions are counted and respect consent. These are hypotheses to test in interviews, not validated willingness-to-pay.

Shopify App Store registration is currently a **one-time $19 per Partner account**. For standard-rate developers, Shopify's revenue share is 0% on the first $1M lifetime gross app revenue earned since 1 Jan 2025 and 15% above it, plus a **2.9% billing processing fee** and applicable taxes; eligibility differs for larger developers. New public apps should use Shopify App Pricing when supported. [S22]

MVP operating estimate [INFERENCE]: roughly $30–$150/month for a small managed app/database, logs, and domain/email at a few pilot stores; $100–$500/month after moderate event volume, depending on hosting and retention. No GPU/LLM is needed; avoid AI inference costs. Legal review is likely the largest upfront expense and varies widely (budget several thousand USD rather than assuming it is free). Security testing, merchant support, and Shopify review preparation are additional costs. Confirm actual regional hosting, taxes/payout availability, and FX/payment costs before committing.

Go-to-market

Pick one niche and make the proof concrete: **Emotional Adaptive Navigation** uses consented interaction patterns to estimate when a shopper may benefit from simpler paths, comparison help, or exploratory suggestions—while leaving the theme's native menu intact. Recruit through Shopify communities, agencies specializing in that vertical, and direct merchant outreach; do not imply Shopify endorsement. Use App Store categories around search/discovery and storefront design only if accurate. Show the merchant configuration, confidence/fallback behavior, consent controls, and evaluation design in the listing. Restrict initial theme support to a small named matrix.

Support burden

Expect support issues around theme conflicts, theme-editor placement, pixel consent not firing, Shopify API version changes, merchant expectations of automatic header replacement, attribution disputes, and "why didn't it increase sales?" Prepare a support guide, store-level kill switch, uninstall cleanup, status page, and transparent experiment report. A solo developer should not promise custom theme rewrites or guaranteed revenue outcomes.

9. Emotional Adaptive Navigation: risks and failure modes (ranked)

Rank	Risk	Likelihood / impact	Mitigation
1	**Affect estimates do not generalize or have unacceptable false positives**	High / High	Separate friction features from affect labels; user-held-out/site-held-out tests; calibrate probabilities; high thresholds and abstention; self-report correction.

Rank	Risk	Likelihood / impact	Mitigation
2	**No paid demand / indistinct from existing personalization**	High / High	Interviews, EAN landing-page and paid-pilot gate; demonstrate incremental value over behavior-only and free Shopify Search & Discovery.
3	**Privacy/consent failure or cross-border noncompliance**	Medium–High / High	Minimize features; consent-aware collection; EU/UK/Pakistan counsel; DPIA where indicated; transfer safeguards; no raw event profiles.
4	**Shopper perceives emotional surveillance or manipulation**	Medium–High / High	Plain-language notice, opt-in/decline, no personal emotion labels in merchant dashboards, no vulnerable-state targeting, no urgency/price changes, easy reset.
5	**Shopify policy violation, especially model training on merchant data**	Medium / High	No pooled model training; scoped merchant opt-in only after legal review; recheck API Terms and maintain deletion rights. [S8]
6	**Theme breakage / native-navigation conflicts**	High / Medium–High	Extension-only; preserve native menu; theme compatibility matrix; preview/rollback and kill switch.
7	**Underpowered A/B test or label leakage leads to false claims**	High / Medium–High	Participant-held-out affect validation; disclose MDE/CI; sales and classifier outcomes reported separately; no post-hoc winner claims.
8	**Store speed regression**	Medium / High	Small asynchronous JS, no blocking requests, static fallback, Lighthouse testing; meet Shopify's ≤10-point delta. [S17, S24]
9	**Multi-tenant leak or vendor breach**	Low–Medium / Very High	Minimize visitor-level storage; tenant isolation/RLS, encryption, least privilege, incident response, subprocessors review, 24-hour Shopify notice.
10	**Accessibility/cultural bias in interaction features**	Medium / Medium–High	Validate by device, language, and accessible input mode; do not use movement norms that penalize assistive technology or infer disability.

10. Emotional Adaptive Navigation: what is realistic versus hype

Doable now for EAN	Needs validation before broad deployment	Avoid / not defensible
A non-biometric, session-only EAN prototype; provisional research scores; calibrated probability hypotheses only after validation; conservative confidence gate and static fallback; optional "How's your search going?" self-report; reversible changes inside an app-owned module; native menus preserved.	Cross-shop affect model accuracy; frustration/confusion estimates from clickstream; user-held-out and locale/device transfer; per-merchant adaptation; whether affect-aware adaptation improves conversion, satisfaction, or task completion over behavior-only personalization; any retained emotion profile.	Claiming accurate or universal emotion detection; diagnosing a shopper or inferring mental-health/vulnerability status; relying on one paper's task accuracy as Shopify accuracy; camera/voice biometrics; changing prices, urgency, eligibility, or suppressing choices based on an inferred state; training across Shopify tenants without consent; guaranteeing sales lift.

11. Additional issues and open product decisions for Emotional Adaptive Navigation

- **Accessibility:** a moving menu can disorient keyboard, screen-reader, cognitive-accessibility, and touch users. Keep native navigation and focus order stable; obey reduced-motion; make change reversible. Accessibility testing is a requirement, not polish.

- **Incentives and conflicting goals:** “conversion” is not always customer welfare. Optimize task completion, satisfaction, and product suitability alongside sales; do not use higher AOV as the only reward.
- **Cold start:** most small stores lack enough interaction data for meaningful recommendations. Use merchant curation, product metadata, and Shopify’s existing recommendations first.
- **Shopify can close the gap:** Search & Discovery, Shop, Sidekick, theme vendors, and recommendation apps can absorb this feature. Build merchant workflow or evidence, not a vague AI moat.
- **Children and sensitive products:** avoid child-directed stores and high-risk categories (health, mental health, financial products) until counsel reviews heightened rules. Minimize retention and avoid sensitive query classification.
- **What must be decided:** one vertical; which themes are supported; what “success” means to merchants; the first release’s supported affect states; which signals need shopper consent; where data is hosted/accessed; and whether a self-report study is worth its added cost.

Naming and marketing claims guide

Keep the product name **Emotional Adaptive Navigation**. Make the descriptor distinguish an affective estimate from a diagnosis.

For the App Store title, Shopify requires a unique, brand-led name; keep **Emotional Adaptive Navigation** intact as the product descriptor (for example, **[Distinctive Brand] — Emotional Adaptive Navigation**), and never imply Shopify endorsement.

Can say (when the shipped behavior supports it)	Must not claim
“Emotion-aware navigation that adapts to how shoppers are interacting with your store.”	“Accurately detects each shopper’s emotion,” “reads their mind,” “knows when they are frustrated,” or clinical/mental-health diagnosis.
“Uses non-biometric, in-session interaction patterns to estimate when navigation help may be useful; shoppers can dismiss or reset suggestions.”	“Guaranteed conversion/AOV lift,” “reduces frustration by X%,” or fabricated/unsubstantiated performance results. Shopify App Store listing rules prohibit statistics/data claims in listing content and images, including verifiable figures.
“Confidence-gated suggestions; native navigation remains available.”	“Bias-free,” “works for every shopper/store,” “the first/only emotion app,” or assurances of legal compliance without substantiation.

Use **“estimate,” “may,” “emotion-aware,”** and **“interaction patterns”** prominently. Claims on the developer website still need evidence under advertising/consumer-protection law. [S13, S17]

Sources and citation key

Provided papers

- **P1** — Sahraoui, A. E. A., Vanderdonckt, J., & Kieffer, S. (2025). *Fractal Adaptive Menus: Concept, Engineering, and Use Case*. EICS Companion ’25, pp. 42–49. DOI: 10.1145/3731406.3734978. Supplied PDF: Fractal Adaptive Menus.pdf.
- **P2** — Gaspar-Figueiredo, D. et al. (2025). *Integrating Human Feedback into a Reinforcement Learning-Based Framework for Adaptive User Interfaces*. EASE 2025; preprint arXiv:2504.20782v1. Supplied PDF: Integrating Human Feedback into a Reinforcement.pdf.
- **P3** — *Eyes Bring out Effectiveness? Applying Eye-Tracking to Explore Consumers’ Visual Perception of Horizontal and Vertical Menu Formats in E-commerce Website Design*. Unreviewed preprint in supplied copy, SSRN 4920862. Supplied PDF: Eyes Bring out Effectiveness.pdf.

Independent research — legal, science, and platform

- **S1** — EU GDPR, Regulation (EU) 2016/679, EUR-Lex official text (adopted 2016; accessed 21 Oct 2026).
- **S2** — UK Data (Use and Access) Act 2025, s.80; UK automated decision provisions in force 5 Feb 2026, legislation.gov.uk (accessed 21 Oct 2026).
- **S3** — UK ICO, *Automated decision-making, including profiling*; latest page update 31 Mar 2026, overview, ADM provisions.

- **S4** — EU ePrivacy Directive 2002/58/EC, Article 5(3), EUR-Lex (adopted 2002; accessed 21 Oct 2026).
- **S5** — UK Data (Use and Access) Act 2025, ss.112 and Schedule 12; PECR amendments effective 5 Feb 2026, s.112, PECR Regulation 6, Schedule A1.
- **S6** — UK ICO, *Guidance on the use of storage and access technologies*, updated 29 Apr 2026, ICO guidance.
- **S7** — Shopify, Web Pixels API, Build web pixels, Pixel privacy, live documentation accessed 21 Oct 2026.
- **S8** — Shopify, API License and Terms of Use, **last updated 7 Oct 2026**; especially §§2.1, 2.3, 6.2, 7.
- **S9** — California Privacy Protection Agency, CCPA regulations/status (current statute/regulations effective 1 Jan 2026; accessed 21 Oct 2026).
- **S10** — CPPA, 2025 ADMT regulations announcement (ADMT significant-decision requirements begin 1 Jan 2027).
- **S11** — Illinois Biometric Information Privacy Act, 740 ILCS 14, Illinois General Assembly (enacted 2008; official page accessed 21 Oct 2026; current text/amendments should be checked by counsel).
- **S12** — EU AI Act, Regulation (EU) 2024/1689, EUR-Lex original; Commission AI Act Service Desk Article 5 and Article 50, consolidated text as at 27 Jul 2026.
- **S13** — FTC, *Bringing Dark Patterns to Light*, Sep 2022, FTC report page; EU Unfair Commercial Practices Directive 2005/29/EC (adopted 2005) and Digital Services Act 2022/2065 (adopted 2022); accessed 21 Oct 2026.
- **S14** — Senate of Pakistan, Personal Data Protection Bill, 2023 status page, record accessed 21 Oct 2026.
- **S15** — Pakistan MoITT, Personal Data Protection Bill, 2023 draft PDF; Senate committee 2024 briefing.
- **S16** — European Commission, GDPR international transfers and adequacy decisions; GDPR Chapter V in S1.
- **S17** — Shopify, App Store requirements, best practices, current live docs accessed 21 Oct 2026.
- **S18** — Shopify, Protected customer data requirements, accessed 21 Oct 2026.
- **S19** — Shopify, Privacy law compliance / mandatory webhooks, privacy requirements, accessed 21 Oct 2026.
- **S20** — Shopify, Theme App Extensions, configuration/restrictions, accessed 21 Oct 2026.
- **S21** — Shopify, Checkout extensions, accessed 21 Oct 2026.
- **S22** — Shopify, App Store revenue share and Shopify App Pricing, accessed 21 Oct 2026.
- **S23** — Shopify, Admin GraphQL menus query, Storefront productRecommendations, and app proxies, API version 2026-10/current docs.
- **S24** — Shopify, app performance guidance, accessed 21 Oct 2026.
- **S25** — Shopify, access scopes, accessed 21 Oct 2026.
- **S26** — Barrett, L.F. et al. (2019). “Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements.” *Psychological Science in the Public Interest*, 20(1), 1–68. DOI: 10.1177/1529100619832930; full text.
- **S27** — Shopify App Store, Shopify Search & Discovery, listing accessed 21 Oct 2026.
- **S28** — Shopify App Store, Nosto; Algolia, listings accessed 21 Oct 2026.
- **S29** — Shopify App Store, LimeSpot; Rebuy (listing was not available at the time retrieved), accessed 21 Oct 2026.
- **S30** — Shopify App Store, Skinalyze; Beautyxia; Meteor Mega Menu, listings accessed 21 Oct 2026.
- **S31** — CPPA, Updated Monetary Thresholds in CCPA, effective 1 Jan 2025; accessed 21 Oct 2026.
- **S32** — Constitution of the Islamic Republic of Pakistan, Article 14, Pakistan Code official PDF (current official copy accessed 21 Oct 2026).
- **S33** — Yamauchi, T. & Xiao, K. (2018). “Reading Emotion From Mouse Cursor Motions: Affective Computing Approach.” *Cognitive Science*, 42(3), 771–819. DOI: 10.1111/cogs.12557. Abstract reports 10-fold CV prediction of 10–20% of variance in positive-affect/attentiveness ratings; N not stated in the accessible abstract.
- **S34** — Freihaut, P. & Göriz, A.S. (2021). “Using the Computer Mouse for Stress Measurement—An Empirical Investigation and Critical Review.” *International Journal of Human–Computer Studies*, 145, 102520. DOI: 10.1016/j.ijhcs.2020.102520. Abstract reports a within-participant lab experiment (N=53) with no clear mouse-use differences under successfully induced high/low stress.
- **S35** — Ghosh, S., Hiware, K., Ganguly, N., Mitra, B., & De, P. (2019). “Emotion Detection from Touch Interactions during Text Entry on Smartphones.” *International Journal of Human–Computer Studies*, 130, 47–57. DOI: 10.1016/j.ijhcs.2019.04.005. Abstract reports a three-week Android field study (N=22), personalized models, and mean AUC-ROC=.73 (SD=.06), maximum .87.

- **S36** — Kołakowska, A. & Landowska, A. (2021). “Keystroke Dynamics Patterns While Writing Positive and Negative Opinions.” *Sensors*, 21(17), 5963. DOI: 10.3390/s21175963; full text.
- **S37** — Lee, P.-M., Tsui, W.-H., & Hsiao, T.-C. (2014). “The Influence of Emotion on Keyboard Typing: An Experimental Study Using Visual Stimuli.” *BioMedical Engineering OnLine*, 13, 81. DOI: 10.1186/1475-925X-13-81; full text.
- **S38** — Lee, P.-M., Tsui, W.-H., & Hsiao, T.-C. (2015). “The Influence of Emotion on Keyboard Typing: An Experimental Study Using Auditory Stimuli.” *PLOS ONE*, 10(6), e0129056. DOI: 10.1371/journal.pone.0129056; full text.
- **S39** — Kelly, C.A. & Sharot, T. (2025; published online 21 Nov 2024). “Web-Browsing Patterns Reflect and Shape Mood and Mental Health.” *Nature Human Behaviour*, 9, 133–146. DOI: 10.1038/s41562-024-02065-6; full text.
- **S40** — Thomas, P. (2014). “Using Interaction Data to Explain Difficulty Navigating Online.” *ACM Transactions on the Web*, 8(4), 1–41. DOI: 10.1145/2656343. The accessible abstract describes web-navigation experiments and difficulty predictors but gives no sample count or accuracy metric.
- **S41** — Yang, L. & Qin, S.-F. (2021). “A Review of Emotion Recognition Methods from Keystroke, Mouse, and Touchscreen Dynamics.” *IEEE Access*, 9, 162197–162213. DOI: 10.1109/ACCESS.2021.3132233; Northumbria research portal. This is a review, not an independent model-validation study.
- **S42** — Epp, C., Lippold, M., & Mandryk, R.L. (2011). “Identifying Emotional States Using Keystroke Dynamics.” *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 715–724. DOI: 10.1145/1978942.1979046. Publisher/abstract metadata describe a field study using experience-sampled states; the accessible abstract does not state N or the validation split. Accuracy descriptions in this report are also attributed to the summary in S41.
- **S43** — Rahman, M.M., Ollington, R., Yeom, S.-J., & Ollington, N. (2024). “Generalisable Sensor-Free Frustration Detection in Online Learning Environments Using Machine Learning.” *User Modeling and User-Adapted Interaction*, 34(4), 1493–1527. DOI: 10.1007/s11257-024-09402-4. The accessible abstract states validation on two independent education datasets but omits N and numerical performance metrics.
- **S44** — El Ayadi, M., Kamel, M.S., & Karray, F. (2011). “Survey on Speech Emotion Recognition: Features, Classification Schemes, and Databases.” *Pattern Recognition*, 44(3), 572–587. DOI: 10.1016/j.patcog.2010.09.020. A methods survey; it does not establish a Shopify storefront accuracy benchmark.

12. Uncertainties and facts not verified

1. **Pakistan law:** the official Senate record reviewed shows the 2023 bill withdrawn from committee and a MoITT draft, but does not prove that no later bill or Gazette notification exists by 21 Oct 2026. Verify with Pakistan counsel and the latest Gazette/Parliament record.
2. **Shopify policies can change rapidly.** The API Terms were updated 7 Oct 2026; the cited App Store and API docs are live versions accessed 21 Oct 2026. Recheck before application submission and before every major data/model change.
3. **UK guidance is still in transition:** the ICO says its post-DUAA ADM guidance (updated 31 Mar 2026) is under consultation. The statutory Act has been checked; final regulator interpretation may change.
4. **UK PECR exception fit:** whether a particular Emotional Adaptive Navigation feature qualifies as website appearance/functionality adaptation, and whether its telemetry is separately regulated, is fact-specific. Do not assume an opt-out exception applies to automated affect inference or interaction data.
5. **Competitor scan is not exhaustive.** App Store search was partially unavailable/incomplete; “no direct emotion-navigation competitor found” is not a verified market-wide absence. Competitor claims and reviews are not independent proof of efficacy.
6. **Paper P3 metadata:** author/date were not visible in the supplied file copy; it is identified as an unreviewed SSRN preprint, SSRN 4920862. Its internal design/reporting inconsistencies limit interpretation.
7. **Shopify theme/menu permissions:** the exact scopes and access approval for reading merchant menus depend on the current API version and app-review decision. MVP should avoid relying on programmatic replacement or writing of menus.
8. **Unit economics and willingness to pay:** the price tiers and hosting costs here are estimates for validation, not market facts. The product has no validated conversion lift or demand yet.
9. **Behavioral-affect evidence coverage:** study tasks, labels, and validation splits differ; Epp et al.’s accessible abstract/review summary does not provide its participant count or held-out-person split, and Rahman et al.’s accessible abstract omits N and numerical metrics. Search reformulation and cart hesitation did not yield a verified commerce affect-classification benchmark in this review. Do not use these evidence gaps as implied proof of accuracy or inaccuracy.